

List DataStructure

```
In [1]: l = []  
l
```

```
Out[1]: []
```

```
In [2]: type(l)
```

```
Out[2]: list
```

```
In [3]: len(l)
```

```
Out[3]: 0
```

```
In [4]: l.append(10)  
l
```

```
Out[4]: [10]
```

```
In [5]: len(l)
```

```
Out[5]: 1
```

```
In [6]: l.append(20,30,40)  
l
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[6], line 1  
----> 1 l.append(20,30,40)  
      2 l  
  
TypeError: list.append() takes exactly one argument (3 given)
```

```
In [7]: l.append(20)  
l.append(30)  
l.append(40)  
l.append(50)  
l
```

```
Out[7]: [10, 20, 30, 40, 50]
```

```
In [9]: len(l)
```

```
Out[9]: 5
```

```
In [10]: l
```

```
Out[10]: [10, 20, 30, 40, 50]
```

```
In [11]: l.append(10)  
l
```

Out[11]: [10, 20, 30, 40, 50, 10]

```
In [12]: l1 = [10,3.5,True,1+2j,'welcome']  
l1
```

Out[12]: [10, 3.5, True, (1+2j), 'welcome']

```
In [13]: l1.append([1,2,3])  
l1
```

Out[13]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [14]: print(l)  
print(l1)
```

[10, 20, 30, 40, 50, 10]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [15]: l
```

Out[15]: [10, 20, 30, 40, 50, 10]

```
In [18]: l3 = l.copy()  
l3
```

Out[18]: [10, 20, 30, 40, 50, 10]

```
In [19]: print(l)  
print(l1)  
print(l3)  
l3
```

[10, 20, 30, 40, 50, 10]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]
[10, 20, 30, 40, 50, 10]

Out[19]: [10, 20, 30, 40, 50, 10]

```
In [20]: l == l3
```

Out[20]: True

```
In [21]: len(l) == len(l3)
```

Out[21]: True

```
In [22]: id(l) == id(l3)
```

Out[22]: False

```
In [24]: id(l) != id(l3)
```

Out[24]: True

```
In [25]: print(id(l))  
print(id(l3))
```

```
1352612723200
1352635366528
```

```
In [26]: l3.clear()
```

```
In [27]: print(l)
         print(l1)
         print(l3)
```

```
[10, 20, 30, 40, 50, 10]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]
[]
```

```
In [28]: del l3
```

```
In [29]: l3
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[29], line 1
----> 1 l3

NameError: name 'l3' is not defined
```

```
In [32]: print(l)
         print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]
```

```
In [33]: l.count()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[33], line 1
----> 1 l.count()

TypeError: list.count() takes exactly one argument (0 given)
```

```
In [35]: l.count(10)
```

```
Out[35]: 2
```

```
In [36]: l.count(20)
```

```
Out[36]: 1
```

```
In [37]: l.count(30)
```

```
Out[37]: 1
```

```
In [39]: print(l)
         print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]
```

```
In [40]: l.extend(l1)
         l1
```

Out[40]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [41]: print(len(l))  
         print(len(l1))
```

12
6

```
In [42]: a = [1,2,3]  
         b = [4,5,6]
```

```
In [43]: a.extend(b)  
         a
```

Out[43]: [1, 2, 3, 4, 5, 6]

```
In [44]: a1 = [1,2,3]  
         b1 = [4,5,6]
```

```
In [45]: a1.extend(b1)  
         a1
```

Out[45]: [1, 2, 3, 4, 5, 6]

```
In [46]: print(a1)  
         print(b1)
```

[1, 2, 3, 4, 5, 6]
[4, 5, 6]

```
In [47]: print(l)  
         print(l1)
```

[10, 20, 30, 40, 50, 10, 10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]
[10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [48]: l.index(20)
```

Out[48]: 1

```
In [49]: l1.index(1+2j)
```

Out[49]: 3

```
In [50]: l1
```

Out[50]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [51]: l1.index('welcome')
```

Out[51]: 4

```
In [52]: l1[:]
```

Out[52]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

```
In [53]: l1[5]
```

Out[53]: [1, 2, 3]

In [54]: l1[4]

Out[54]: 'welcome'

In [55]: l1

Out[55]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [56]: l1[:3]

Out[56]: [10, 3.5, True]

In [57]: l1

Out[57]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [58]: l1[3:]

Out[58]: [(1+2j), 'welcome', [1, 2, 3]]

In [59]: l1

Out[59]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [60]: l1[0:5:2]

Out[60]: [10, True, 'welcome']

In [61]: l1[-3]

Out[61]: (1+2j)

In [62]: l1[-3:]

Out[62]: [(1+2j), 'welcome', [1, 2, 3]]

In [63]: l1[::-1]

Out[63]: [[1, 2, 3], 'welcome', (1+2j), True, 3.5, 10]

In [64]: l1

Out[64]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [65]: l2[::-2]

```
-----
NameError                                Traceback (most recent call last)
Cell In[65], line 1
----> 1 l2[::-2]

NameError: name 'l2' is not defined
```

In [66]: l1

Out[66]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [67]: l1[4]

Out[67]: 'welcome'

In [68]: print(l1[4][0])
print(l1[4][1])
print(l1[4][2])
print(l1[4][3])
print(l1[4][4])
print(l1[4][5])
print(l1[4][6])
print(l1[4][7])

w
e
l
c
o
m
e

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[68], line 8  
      4 print(l1[4][3])  
      5 print(l1[4][4])  
      6 print(l1[4][5])  
      7 print(l1[4][6])  
----> 8 print(l1[4][7])  
  
IndexError: string index out of range
```

In [69]: l1

Out[69]: [10, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In [71]: l1[0]

Out[71]: 10

In [72]: l1[0] = 100

In [73]: l1

Out[73]: [100, 3.5, True, (1+2j), 'welcome', [1, 2, 3]]

In []: